

- Description

This module manages a smart gas stove flame control system, ensuring that:

1. The system continuously monitors the current flame intensity (in arbitrary units).
2. If the flame intensity falls below a predefined minimum threshold, the system increases the gas flow to boost the flame.
3. If the flame intensity exceeds a predefined maximum threshold, the system decreases the gas flow to reduce the flame intensity.
4. If the flame intensity is within the optimal range, no adjustment is made.

This system helps maintain an optimal flame level for efficient cooking and safety.

- Specification

module SmartGasStoveControl

var currentFlameIntensity, minFlameIntensity, maxFlameIntensity: real

var gasFlowRate: real

process updateFlameIntensity(newIntensity: real) returns (currentFlameIntensity: real, gasFlowRate: real)

functional scenario form:

currentFlameIntensity < minFlameIntensity && gasFlowRate = gasFlowRate * 1.2, output "ALERT: Flame intensity (" + currentFlameIntensity + ") is too low. Increasing gas flow to " + gasFlowRate + " L/min.", Gas flow increased by 20%.

||

currentFlameIntensity > maxFlameIntensity && gasFlowRate = gasFlowRate * 0.8, output "ALERT: Flame intensity (" + currentFlameIntensity + ") is too high. Decreasing gas flow to " + gasFlowRate + " L/min.", Gas flow decreased by 20%.

||

currentFlameIntensity >= minFlameIntensity && output "Flame intensity (" + currentFlameIntensity + ") is optimal. Gas flow remains at " + gasFlowRate + " L/min.", No adjustment to gas flow.

end process

end module

- Java Code

```
package code;
public class c100_SmartGasStove {
    private double minFlameIntensity;
    private double maxFlameIntensity;
    private double currentFlameIntensity;
    private double gasFlowRate;
    public c100_SmartGasStove(double minFlameIntensity, double maxFlameIntensity, double initialFlameIntensity, double initialGasFlowRate) {
        this.minFlameIntensity = minFlameIntensity;
        this.maxFlameIntensity = maxFlameIntensity;
        this.currentFlameIntensity = initialFlameIntensity;
    }
}
```

```

        this.gasFlowRate = initialGasFlowRate;
    }
    public void updateFlameIntensity(double newFlameIntensity) {
        currentFlameIntensity = newFlameIntensity;
        adjustGasFlow();
        System.out.printf("Current Flame Intensity: %.2f, Gas Flow Rate
: %.2f L/min%n",
                        currentFlameIntensity, gasFlowRate);
        if (currentFlameIntensity < minFlameIntensity) {
            System.out.println("ALERT: Flame intensity too low! Increasing gas flow.");
        } else if (currentFlameIntensity > maxFlameIntensity) {
            System.out.println("ALERT: Flame intensity too high! Decreasing gas flow.");
        } else {
            System.out.println("Flame intensity is optimal. No adjustment needed.");
        }
    }
    private void adjustGasFlow() {
        if (currentFlameIntensity < minFlameIntensity) {
            gasFlowRate = gasFlowRate * 1.2;
        } else if (currentFlameIntensity > maxFlameIntensity) {
            gasFlowRate = gasFlowRate * 0.8;
        }
    }
    public double getCurrentFlameIntensity() {
        return currentFlameIntensity;
    }
    public double getGasFlowRate() {
        return gasFlowRate;
    }
    public static void main(String[] args) {
        c100_SmartGasStove stove = new c100_SmartGasStove(50.0, 80.0, 70.0, 1.0);

        // testcase-VT:
        //     stove.updateFlameIntensity(45.0);
        //     stove.updateFlameIntensity(85.0);
        //     stove.updateFlameIntensity(65.0);

        // testcase-FT:
        //     stove.updateFlameIntensity(-1.7976931348623157E308);
        //     stove.updateFlameIntensity(0.0);

```

```
//      stove.updateFlameIntensity(-4.9E-324);

//      // testcase-Z3:
//      stove.updateFlameIntensity(45.0);
//      stove.updateFlameIntensity(85.0);
//      stove.updateFlameIntensity(65.0);

//      // testcase-UVT:
//      stove.updateFlameIntensity(45.0);
//      stove.updateFlameIntensity(85.0);
//      stove.updateFlameIntensity(65.0);
    }
}
```

- Coverage report

1. Test case number: 3
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	834	842
▼ src	1.0 %	8	834	842
▼ code	1.0 %	8	834	842
▼ c100_SmartGasStove.java	100.0 %	8	0	8
▼ c100_SmartGasStove	100.0 %	8	0	8
● updateFlameIntensity(double)	100.0 %	4	0	4
■ adjustGasFlow()	100.0 %	4	0	4
● c100_SmartGasStove(double, dc		0	0	0
● main(String[])		0	0	0

3. Vibration testing instruction coverage: 94.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.5 %	106	21,716	21,822
▼ src	0.5 %	106	21,716	21,822
▼ code	0.5 %	106	21,716	21,822
▼ c100_SmartGasStove.java	94.6 %	106	6	112
▼ c100_SmartGasStove	94.6 %	106	6	112
● updateFlameIntensity(double)	100.0 %	47	0	47
■ adjustGasFlow()	100.0 %	26	0	26
● c100_SmartGasStove(double, dc	100.0 %	15	0	15
● main(String[])	100.0 %	18	0	18

4. Fuzz testing branch coverage: 25%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.2 %	2	840	842
▼ src	0.2 %	2	840	842
▼ code	0.2 %	2	840	842
▼ c100_SmartGasStove.java	25.0 %	2	6	8
▼ c100_SmartGasStove	25.0 %	2	6	8
● updateFlameIntensity(double)	25.0 %	1	3	4
■ adjustGasFlow()	25.0 %	1	3	4
● c100_SmartGasStove(double, dc		0	0	0
● main(String[])		0	0	0

5. Fuzz testing instruction coverage: 72.3%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	81	21,741	21,822
▼ src	0.4 %	81	21,741	21,822
▼ code	0.4 %	81	21,741	21,822
▼ c100_SmartGasStove.java	72.3 %	81	31	112
▼ c100_SmartGasStove	72.3 %	81	31	112
● updateFlameIntensity(double)	72.3 %	34	13	47
■ adjustGasFlow()	53.8 %	14	12	26
● c100_SmartGasStove(double, dc	100.0 %	15	0	15
● main(String[])	100.0 %	18	0	18

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	834	842
▼ src	1.0 %	8	834	842
▼ code	1.0 %	8	834	842
▼ c100_SmartGasStove.java	100.0 %	8	0	8
▼ c100_SmartGasStove	100.0 %	8	0	8
● updateFlameIntensity(double)	100.0 %	4	0	4
■ adjustGasFlow()	100.0 %	4	0	4
● c100_SmartGasStove(double, dc		0	0	0
● main(String[])		0	0	0

7. Z3 instruction coverage: 94.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.5 %	106	21,716	21,822
▼ src	0.5 %	106	21,716	21,822
▼ code	0.5 %	106	21,716	21,822
▼ c100_SmartGasStove.java	94.6 %	106	6	112
▼ c100_SmartGasStove	94.6 %	106	6	112
● updateFlameIntensity(double)	100.0 %	47	0	47
■ adjustGasFlow()	100.0 %	26	0	26
● c100_SmartGasStove(double, dc	100.0 %	15	0	15
● main(String[])	100.0 %	18	0	18

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	834	842
▼ src	1.0 %	8	834	842
▼ code	1.0 %	8	834	842
▼ c100_SmartGasStove.java	100.0 %	8	0	8
▼ c100_SmartGasStove	100.0 %	8	0	8
● updateFlameIntensity(double)	100.0 %	4	0	4
■ adjustGasFlow()	100.0 %	4	0	4
● c100_SmartGasStove(double, dc		0	0	0
● main(String[])		0	0	0

9. Unimproved V-method instruction coverage: 94.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.5 %	106	21,716	21,822
▼ src	0.5 %	106	21,716	21,822
▼ code	0.5 %	106	21,716	21,822
▼ c100_SmartGasStove.java	94.6 %	106	6	112
▼ c100_SmartGasStove	94.6 %	106	6	112
● updateFlameIntensity(double)	100.0 %	47	0	47
■ adjustGasFlow()	100.0 %	26	0	26
● c100_SmartGasStove(double, dc	100.0 %	15	0	15
● main(String[])	100.0 %	18	0	18